

BUS Tracking System

(Design Engineering project)

So as per the requirement I have chosen bus tracking system because in my semester we have made a project related to it and have continued the project throughout the semesters. Now as per the lecture and task given it was said to make your own notes about AWS services.

Q1. AWS Cloud Computing:

Cloud=Internet + storing something online + services provided.

Example was Netflix, Instagram, WhatsApp backup.

Instead of buying physical servers which will be not useful during night as bus don't work at night time, the school "rents" computing power only when buses are moving (e.g., 6 AM–8 AM and 2 PM–4 PM). This allows the system to automatically handle thousands of GPS pings during rush hour without crashing.

Cloud service model:

Bus tracking System uses a mix: **IaaS** for the core map server, **PaaS** for the real-time data processing (Lambda), and **SaaS** for the parent mobile app interface.

Q.2 Amazon EC2 (Elastic Compute Cloud):

It is basically cloud computer or virtual machine type where we can have access to let's say MacBook, we want to buy but cannot afford, here comes EC2 we can rent similar system like MacBook.

Bus tracking system:

You run a web server on **EC2** that hosts the dispatcher's map interface. During the night, you spin up a powerful EC2 to analyse the day's GPS data, optimize routes for fuel efficiency, and generate reports. You only pay for the 9 hours (if we assume) it was running. If you left it on 24/7, you'd be paying for a supercomputer sitting idle when no one is using it.

Q.3 AWS Lambda (Serverless Compute):

Heart of AWS. When request sent AWS code will execute and code run until service provided and then it will not run.

When a bus GPS sends a "Stop Approaching" signal, **Lambda** triggers instantly to calculate the ETA and push an SMS to parents. Since buses only send data at intervals, you pay only for the milliseconds the code runs, not for a server sitting idle 24/7. This all the developers will be using as it reduces cost and serverless.

Q.4 Amazon S3 (Simple Storage Service):

To store any amount of data. subpoints were Bucket, object, Standard class, Glacier class. Every GPS ping, driver log, and maintenance photo is stored into **S3** buckets. If your bus has cameras video recording can be stored, driver photo, maintenance book etc.. can be stored.

Q.5 AWS storage suite:

Amazon s3, Amazon EBS, Amazon EFS, Amazon FSX, we learned about it.in bus tracking system. Amazon EBS is a block storage service attached to an EC2 instance. In a bus tracking system, it is used to store the application, operating system, and database containing GPS locations, bus details, and user information. Whereas Amazon EFS stores Shared files and documents.

Q.6 Amazon Database:

We learned about Amazon RDS (Relational database) and Amazon DynamoDB. Stores the real-time location of 500+ buses. Stores structured data like Student IDs, Bus Driver Names, Route Schedules, and Billing Information. basically it gives instant read access for the live map and handles complex queries like "Show all students on Route from KH6 -to samarpan circle."

Q.7 AWS Security Services:

We saw 2 sub points Amazon inspector and Amazon guard Duty. These services ensure that the bus tracking application remains secure, reliable, and compliant while protecting user and location data. Secure login for users and administrators. Protects passenger and GPS location data. Prevents cyberattacks such as DDoS. Amazon guard Duty detects security threats.

Q.8 IAM (Identity and Access Management):

Role is defined like who will have access to which services. Let's say user would have which services whereas developers would have which accesses or would work on which services.

- **Dispatcher:** Can only read the live map and send messages
- **Driver App:** Can *only* send GPS data, not read student addresses.
- **Admin:** Full access to billing and user management.
- **Parent App:** can only see their child's locations on the app and not the others child's location.

Q.9 Real World Architecture: Web application

At last, we saw about load balancer, user, EC2 server, Amazon RDS. Load balancer distribute incoming requests(many) across multiple server to prevent overloading.